

DUNKERQUE, France

WMO: 070100

Lat: 51.0558N

Lon: 2.3394E

Elev: 52

StdP: 14.67

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	26.2	29.4	18.8	14.2	30.7	22.3	16.8	31.0	35.5	45.8	32.7	45.3	12.6	80	0.661

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	9.7	77.4	66.0	74.1	65.2	71.6	64.1	68.3	75.0	66.8	72.2	65.3	70.1	10.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.8	95.5	71.7	64.5	91.0	70.2	63.1	86.6	68.7	32.6	75.2	31.4	72.3	30.2	70.3	74.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 30.1	27.1	24.5	23.9	89.8	3.3	5.7	21.5	93.8	19.5	97.1	17.7	100.3	15.3	104.4		
(5)			22.0	71.6	4.3	1.4	19.0	72.6	16.4	73.4	14.0	74.2	10.9	75.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.0	42.0	42.5	45.7	50.4	55.8	60.8	64.8	65.5	61.4	55.2	48.4	43.2
	DBStd	9.81	6.54	5.92	4.93	4.95	4.72	4.31	3.90	3.61	3.93	5.12	5.37	6.43
	HDD50	1001	254	215	147	53	6	0	0	0	0	15	89	222
	HDD65	4497	714	631	598	439	288	141	51	34	122	304	498	677
	CDD50	2105	5	5	14	64	186	323	457	481	341	178	41	10
	CDD65	125	0	0	0	0	3	14	43	50	13	2	0	0
	CDH74	338	0	0	0	1	18	59	100	123	34	3	0	0
	CDH80	84	0	0	0	0	4	17	27	29	7	0	0	0
	WSAvg	12.7	14.6	14.2	13.4	12.2	12.3	11.8	11.5	11.4	12.0	12.5	13.1	14.0
	PrecAvg	27.0	2.3	1.5	2.1	1.8	2.0	2.0	2.1	2.0	2.5	2.9	3.3	2.4
(15) Precipitation	PrecMax	34.1	5.4	3.2	4.4	5.1	6.1	4.9	5.4	3.9	5.6	7.9	7.6	4.9
	PrecMin	20.0	0.4	0.2	0.3	0.2	0.4	0.0	0.4	0.2	0.2	0.5	1.4	0.7
	PrecStd	4.1	1.1	0.8	1.1	1.0	1.3	1.2	1.2	1.0	1.5	1.8	1.7	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	58.0	63.8	70.4	75.8	81.8	83.0	84.9	77.9	72.2	62.0	57.7
		MCWB	52.6	50.6	54.5	57.0	61.1	66.3	67.4	68.9	64.8	62.8	57.8	54.2
	2%	DB	54.2	53.9	57.1	64.1	68.5	73.4	76.0	77.2	72.6	66.5	59.5	55.9
		MCWB	51.5	49.9	51.3	54.5	59.1	64.5	66.4	66.7	63.8	60.9	56.1	53.0
	5%	DB	52.0	52.0	54.0	59.9	64.9	69.6	73.0	73.5	69.5	63.8	57.7	54.0
		MCWB	49.3	48.7	49.8	53.1	57.9	62.4	65.1	66.0	62.6	59.4	54.6	51.3
	10%	DB	50.1	50.4	52.0	56.8	62.2	67.0	70.6	71.1	67.2	61.8	55.6	51.9
		MCWB	47.5	47.4	48.6	51.5	56.4	60.7	63.8	64.4	61.4	57.9	52.8	49.1
	0.4%	WB	53.2	52.6	54.9	57.9	63.0	68.2	70.4	70.7	67.7	63.7	58.5	55.3
		MCDB	55.1	56.0	60.4	67.2	70.9	78.5	78.8	80.5	74.3	69.2	60.4	57.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	51.7	50.7	52.1	55.6	60.2	65.2	68.2	68.5	65.6	61.7	56.7	53.4
		MCDB	53.9	53.2	56.0	62.2	66.4	71.5	74.5	74.8	70.1	65.9	59.0	55.5
	5%	WB	49.6	49.2	50.5	53.7	58.7	63.2	66.4	67.1	63.8	60.1	55.0	51.6
		MCDB	51.7	51.6	53.5	58.8	64.0	68.4	71.6	72.1	67.9	63.1	57.3	53.9
	10%	WB	47.7	47.6	49.1	52.1	57.2	61.5	64.9	65.5	62.4	58.6	53.0	49.4
		MCDB	49.9	50.1	51.8	56.2	61.4	66.1	69.4	70.1	66.4	61.5	55.5	51.7
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.2	8.0	9.4	9.2	9.4	9.5	9.7	9.4	8.9	7.0	6.8
		MCDBR	7.8	9.3	11.5	15.4	14.5	14.9	14.1	14.1	14.0	11.3	7.8	7.8
		MCWBR	7.5	7.1	7.6	8.3	7.8	8.5	7.6	7.5	7.4	7.4	6.8	7.5
	5% WB	MCDBR	7.7	8.2	10.2	13.6	12.7	13.6	12.2	12.3	11.7	10.3	7.5	7.7
		MCWBR	7.7	7.0	7.4	7.8	7.3	8.3	7.4	7.1	7.3	7.6	7.1	7.6
(40) Clear-Sky Solar Irradiance	taub		0.335	0.354	0.393	0.410	0.420	0.419	0.418	0.409	0.388	0.371	0.346	0.329
	taud		2.366	2.348	2.243	2.227	2.255	2.279	2.290	2.333	2.380	2.395	2.380	2.350
	Ebn at Noon		214	242	252	262	265	266	264	260	253	233	207	195
	Edh at Noon		21	27	36	40	41	40	39	36	31	25	20	18
(44) All-Sky Solar Radiation	RadAvg		249	471	855	1384	1621	1768	1675	1423	1059	608	292	192
	RadStd		22	84	108	131	124	145	164	119	73	53	31	24

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (71 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page